ITA0608-MACHINE LEARNING

LAB PROGRAMS

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EXP NO:2 Candidate-Elimination Algorithm

AIM:

To implement and demonstrate the Candidate -Elimination algorithm for finding the most specific hypothesis from the given training data

PROCEDURE:

1. Define the training dataset
2. Initialize the hypothesis with the most specific values (0).
3. Consider only the positive training examples
4. Update the hypothesis by replacing mismatched attributes with ?
5. Display the final hypothesis

PROGRAM:

1.

```
import pandas as pd

data = pd.DataFrame({
  'Outlook':['Sunny','Sunny','Rainy','Sunny','Overcast','Rainy'],
  'Temp':['Hot','Hot','Mild','Cool','Hot','Cool'],
  'Humidity':['High','High','High','Normal','Normal','Normal'],
  'Wind':['Weak','Strong','Weak','Weak','Strong','Strong'],
  'Play':['Yes','Yes','No','Yes','Yes','No']
})

X = data.iloc[:,:-1].values
Y = data.iloc[:,-1].values

S = ['0'] * len(X[0])
G = ['?'] * len(X[0])

for i in range(len(Y)):
  if Y[i] == "Yes":
    for j in range(len(S)):
      if S[j] == '0':
        S[j] = X[i][j]
      elif S[j] != X[i][j]:
        S[j] = '?'
  else:
    for j in range(len(G)):
      if S[j] != X[i][j]:
        G[j] = S[j]

print("Specific Hypothesis:", S)
```

```
print("General Hypothesis:", G)
```

OUTPUT:

```
Specific Hypothesis: ['?', '?', '?', '?']
General Hypothesis: ['?', '?', '?', '?']
```

2.

PROGRAM:

```
import pandas as pd

data = pd.DataFrame({
  'Study':['High','Medium','Low','High','Medium'],
  'Attendance':['Good','Good','Poor','Good','Average'],
  'Assignment':['Yes','Yes','No','Yes','Yes'],
  'Pass':['Yes','Yes','No','Yes','Yes']
})

X = data.iloc[:,:-1].values
Y = data.iloc[:,-1].values

S=['0']*len(X[0])
G=['?']*len(X[0])

for i in range(len(Y)):
  if Y[i]=='Yes':
    for j in range(len(S)):
      if S[j]=='0':
        S[j]=X[i][j]
      elif S[j]!=X[i][j]:
        S[j]='?'
  else:
    for j in range(len(G)):
      if S[j]!=X[i][j]:
        G[j]=S[j]

print("S =",S)
print("G =",G)
```

OUTPUT:

```
S = ['?', '?', 'Yes']
G = ['?', 'Good', 'Yes']
```

3.

PROGRAM:

```
import pandas as pd

data = pd.DataFrame({
  'Income':['High','Medium','Low','High','Medium'],
  'Credit':['Good','Good','Poor','Excellent','Good'],
  'Job':['Yes','Yes','No','Yes','Yes'],
  'Loan':['Yes','Yes','No','Yes','Yes']
})

X=data.iloc[:,:-1].values
Y=data.iloc[:,-1].values

S=['0']*len(X[0])
G=['?']*len(X[0])

for i in range(len(Y)):
  if Y[i]=='Yes':
    for j in range(len(S)):
      if S[j]=='0':
        S[j]=X[i][j]
      elif S[j]!=X[i][j]:
        S[j]='?'
  else:
    for j in range(len(G)):
      if S[j]!=X[i][j]:
        G[j]=S[j]

print("S =",S)
print("G =",G)
```

OUTPUT:

```
S = ['?', '?', 'Yes']
G = ['?', 'Good', 'Yes']
```

4.

PROGRAM:

```
import pandas as pd

data = pd.DataFrame({
  'Experience':['High','Medium','Low','High','Medium'],
  'Performance':['Good','Good','Poor','Excellent','Good'],
```

```
    'Training':['Yes','Yes','No','Yes','Yes'],
    'Promotion':['Yes','Yes','No','Yes','Yes']
})

X=data.iloc[:,:-1].values
Y=data.iloc[:,-1].values

S=['0']*len(X[0])
G=['?']*len(X[0])

for i in range(len(Y)):
    if Y[i]=='Yes':
        for j in range(len(S)):
            if S[j]=='0':
                S[j]=X[i][j]
            elif S[j]!=X[i][j]:
                S[j]='?'
    else:
        for j in range(len(G)):
            if S[j]!=X[i][j]:
                G[j]=S[j]

print("S =",S)
print("G =",G)
```

OUTPUT:

```
S = ['?', '?', 'Yes']
G = ['?', 'Good', 'Yes']
```

5.

PROGRAM:

```
import pandas as pd

data = pd.DataFrame({
    'Age':['Young','Young','Old','Middle','Young'],
    'Income':['High','Medium','Low','Medium','High'],
    'Member':['Yes','Yes','No','Yes','Yes'],
    'Purchase':['Yes','Yes','No','Yes','Yes']
})

X=data.iloc[:,:-1].values
Y=data.iloc[:,-1].values

S=['0']*len(X[0])
G=['?']*len(X[0])

for i in range(len(Y)):
```

```
    if Y[i]=='Yes':
        for j in range(len(S)):
            if S[j]=='0':
                S[j]=X[i][j]
            elif S[j]!=X[i][j]:
                S[j]='?'
    else:
        for j in range(len(G)):
            if S[j]!=X[i][j]:
                G[j]=S[j]

print("S =",S)
print("G =",G)
```

OUTPUT:

```
S = ['?', '?', 'Yes']
G = ['Young', '?', 'Yes']
```

RESULT :

Thus ,the Candidate-Elimination algorithm was implemented successfully ,and the most specific hypothesis was obtained .

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